

MuscleGrid — Serial & Label System

Master Plan & Standard Operating Procedure · v1 · 13 July 2026

The single source of truth for how serial numbers are formed, how every label prints, in what order, and the guards that keep it correct — forever.

1. The Serial Number — one format, always

Every finished unit gets ONE serial in this exact shape:

MG · LIB · 2607 · 00042

Part	Means	Example
MG	MuscleGrid brand prefix (fixed)	MG
FAMILY (3 letters)	The item type — coded so you can read the product from the serial itself	LIB
YYMM (4 digits)	Year + Month of creation	2607 = July 2026
SEQ (5 digits)	Atomic running number — unique, never repeats	00042

So **MGLIB260700042** reads as: MuscleGrid · Lithium Battery · July 2026 · unit #42. The item type and month are encoded right in the number.

Family codes (the "item type")

Code	Product family	Code	Product family
LIB	Lithium Battery	SOL	Solar Panel
INV	Inverter	CTL	MPPT / Controller
STB	Stabilizer	EVC	EV Charger
PCB	PCB / Board	—	else: falls back to SKU code

Why it can never mislabel the family: the family is chosen from the product *type* first, then a battery-aware classifier. A "Battery-Less" *inverter* is correctly read as INV — it is NOT misread as a battery just because its name contains the word "battery". This was a real bug; it is now fixed and guarded.

2. How a serial is created

Stage	Who / where	What happens
Build	Angad (battery) / Gaurav (inverter) — Build Unit panel	Scan the raw material — battery: BMS ; inverter: motherboard + WiFi board . The serial is minted (<code>in_stock</code> , <code>origin=built</code>), the components are linked to it (genealogy), and the serial label prints.
Stock intake	Add-to-stock mode	Scan the serial already on a finished unit → recorded <code>in_stock</code> .
Dispatch	Ship Desk pack	FIFO : consume the OLDEST in-stock serial <i>for that exact product</i> ; if none exist, mint a fresh one.

The single minting function: every serial across build, stock and dispatch is created by ONE function (`_generate_serial`) using ONE family map (`_serial_family_prefix`). There is no second code path — so the format can never drift.

3. The guards — why a serial is never printed incorrectly

#	Guard	What it prevents
1	Family from product-type + battery-aware classifier	An inverter getting a battery (LIB) serial
2	FIFO matches on the product's SKU link only	A wrong product's serial being handed to an order — structurally impossible
3	Labels read the product name live from the SKU link	A stale/mistyped stored name showing the wrong product on a label
4	Serial-label idempotency (<code>label_printed_at</code>)	The same physical serial label printing twice
5	Component single-use lock	One BMS / board being built into two different units
6	Atomic sequence counter + DB-unique serial	Two units ever sharing a serial number
7	Cross-category Amazon SKU-map guard	A battery listing auto-mapped to an inverter (or charger)
8	Scan-the-unit-serial at pack (recommended mandatory)	The label serial differing from the physical unit in the box

4. The labels — what each one carries

Every serial-bearing label now has a **scannable Code128 barcode** of the serial, so it can be scanned at packing and at the gate.

Label	Goes	Printer	Carries
Serial label	on the unit	Thermal	Product (live) · Serial + Code128 barcode · verify QR · component genealogy (BMS / MB / WiFi) · Mfg date · stock/new flag
MRP label	box exterior	Thermal	Product (full name) · M.R.P. ₹ · Serial + Code128 barcode · Net qty · Mfg · Origin
QC label	box exterior	Thermal	Order ref · Ship-to · Contents · S/N + Code128 barcode · QC-passed seal · Packed-by · verify QR
Care card	in the box	Thermal	"Specially crafted for {customer}" · Product · Warranty · WiFi ID (inverters) · Care tip · QR
Shipping label	box exterior	Thermal/A4	AWB + Code128 barcode · Deliver-to (name, address, phone) · From (MuscleGrid) · Order / Weight / Prepaid-COD
Tax invoice	in the box	A4	GST invoice — firm, GSTIN, items, HSN, CGST/SGST or IGST, total
Packing slip	on/in box	A4	Amazon-format: ship-to, order summary, items, declaration

5. The print SEQUENCE — fixed, everywhere

Packing fires from ONE place (`print_pack_set`), always in this order — A4 documents first, then the thermal labels:

- ❶ Courier / shipping label (A4) ❷ Tax invoice (A4) ❸ Packing slip (A4)
- ❹ Serial label (thermal) ❺ QC label (thermal) ❻ MRP label (thermal) ❽ Care card (thermal)

Because all packing goes through this one function, the order is identical for every order, every operator, every time. A **duplicate-print guard** (`packed_at`) blocks a second full print unless someone explicitly chooses "Reprint".

6. How the format can never drift away

- **One minting function** for all serials → one format everywhere (build, stock, dispatch).
- **One family map** → item-type coding is consistent and battery-aware.
- **Labels read product live from the SKU link** → renaming a SKU never mislabels an old serial.
- **One packing function** → the label set and its print order are the same for everyone.
- **Guards 1–8** above are in code, not in operator memory — they hold even if a step is done in a hurry.

7. Operator SOP (the floor)

Step	Screen	Action
Build a battery	Build Unit	Angad: pick model → scan BMS → serial prints → stick on unit
Build an inverter	Build Unit	Gaurav: pick model → scan motherboard + WiFi → serial prints → stick on unit
Record stock	Build → Add to stock	Pick product → Single / Multiple → scan each serial in
Pack an order	Ship Desk → Print pack	Scan the unit's serial → the 7-label set prints in order
Dispatch out	Gate → Outward	Pick courier once → scan every AWB (barcode) → out

8. Monthly audit checklist

- **No duplicate serials** — every serial number is unique (DB-enforced).
- **No stale names** — each serial's product matches its SKU link (auto-synced; last full repair fixed 757 of 1,089).
- **Family correct** — inverters read `INV`, batteries `LIB`, stabilizers `STB`.
- **FIFO health** — oldest in-stock serial per product is consumed first; no "new" minted while stock exists.
- **Barcodes scan** — serial / MRP / QC / shipping barcodes read cleanly on the handheld.
- **Print order** — spot-check a packed order: 7 labels present, thermal set in the fixed sequence.

This document is the agreed standard. Any change to the serial format, a label's content, or the print order must update this plan first, then the code — never the other way around.